

Proposed SOLARIS (systemic or localized attribution of perfusion causing ischemic skin injury) Clinical Assessment Tool

Section 1: Introducing the tool:

1. Failure” occurs when the skin and underlying tissue atrophy/die due to “hypoperfusion”, or inadequate blood flow during critical illness, chronic disease progression, or the dying process. Unlike pressure injuries caused by external mechanical forces, skin failure results from “internal vascular compromise” affecting tissue perfusion. Skin failure may often develop over “bony prominences”, but the primary etiology refers to a systemic cause of ischemia rather than localized ischemia due to mechanical forces as seen in pressure injuries. This spectrum of skin changes may include:
 - a. Acute Skin Failure (ASF): critical illness-related (sepsis, shock, MODS)
 - b. Chronic Skin Failure: Progressive decline from chronic diseases (CHF, ESRD, Cirrhosis), reperfusion injuries.
 - c. End-Stage Skin Failure / Skin changes at the end of life (SCALE): natural skin changes towards the end of the dying process (Kennedy Terminal Ulcer / KTU, TB-TTI, Unavoidable pressure injuries)
2. This tool utilizes an initial stop-gap to determine if pressure forces vs systemic hypoperfusion were likely attributed etiologies, followed by a
3. This tool serves to guide clinicians in determining a differential for wound etiology, and provides a systematic approach to identify, differentiate, document, and manage skin manifestations from other etiologies utilizing evidenced-based criteria **adapted from Hill/Peterson, Latimer, PAWSIC, and clinical practice. It does not take the place of a provider, or final determination of legality/liability.**

Section 2: **Stop-gap** reduce risk for false positives, or over-use of skin failure as primary dx

- a. Standard of Care Met / Prevention Measures Documented within 72 hours PRIOR to wound onset at least Q shift
 - i. Braden score with risk based interventions
 - ii. Turning and repositioning schedule per protocol

iii.Specialty Support in use, if applicable

iv.Offloading device (boots, wedge depending on applicable wound location/locations)

v.Moisture management per protocol

vi.Nutritional support per protocol

STOP-GAP Scoring:

Tally Range	Point Attribution	Determination
≥ 5/6	3 points	S.O.C. (standard of care) met, proceed to Domain 2
≥ 3/6	1 point	S.O.C. partially met, document gaps in care and corrective actions. Proceed to Domain 2
< 3/6	0 points	S.O.C. not met, proceed to Domain 2. • If D1+D2 ≥8, gap is acknowledged with strong support for skin failure despite likely pressure component, classify as unavoidable pressure injury (UPI) • If D1+D2 <8, classify as avoidable pressure injury (API) due as there is insufficient evidence to cover gaps. May note SF as secondary factor/dx.
Total: ___/3		

Section 3:

1. Identifying Risk

a. Domain 1: Clinical Context Supporting Critical illness, **SATA**

i.Sub-Domain: Acute Dx contributing to **systemic** hypoperfusion

1. Sepsis, septic shock
2. Cardiogenic shock
3. Hypovolemic shock
4. Cardiac arrest (within 7 days)
5. AKI requiring CRRT/HD
6. Acute Heart Failure
7. Hepatic Failure
8. DIC/ acute coagulopathy
9. Anoxic Brain Injury
- 10.COVID-19 with severe systemic interactions
- 11.Acute hypoxemic respiratory failure
- 12.Other acute hypoperfusion status:_____

Sub-Domain Scoring	Attributed Point Range
Z> or = to 3 acute Dx (severe MODS)	3
2 acute Dx (moderate MODS)	2
1 Acute Dx (not MODS)	1
0 Acute Dx	0
Total Sub-Domain Scoring (D1A)___/3	

ii. Sub-Domain: Acute on chronic dx / Dx with decompensation **SATA**

1. End stage HF (repeated admissions, diuretic resistance)
2. ESRD **requiring HD**
3. Advanced cirrhosis
4. Metastatic Cancer with organ involvement
5. Advanced COPD with respiratory failure
6. Severe peripheral arterial disease
7. Advanced DM with end-organ damage
8. Chronic malnutrition / failure to thrive
9. Other chronic hypoperfusion state: ____

Sub-Domain Scoring	Attributed Point Range
Z> or =2 chronic conditions with current decomposition OR cumulative organ damage	2
1 chronic condition	1
0	0
Total Sub Domain Scoring (D1b)___/2	

iii. Sub-Domain: Physician-end documentation SATA

1. Documented as critically ill with increased risk for morbidity/mortality?
2. Patient has systemic hypoperfusion OR documentation consistent with hypoperfusion as a clinical concern (shock, hemodynamic instability, end-organ hypoperfusion)
3. Patient is nearing end of life / has grave or poor prognosis
4. Provider acknowledges that skin changes may be related to systemic illness (non pressure etiology).

Sub-Domain Scoring	Attributed Point Range
Z> or =2 documented elements present	2
1 documented element	1
0 documented elements	0
Total Domain (D1a+b+c): __/7	

**** Finer Points:** Including provider feedback and documentation fosters multidisciplinary approach and interdisciplinary communication. Provider documentation should **support**, **not** drive.

b. Domain 2: Hypoperfusion Markers macro vs micro circulation **0-4**

i.Sub-Domain: Laboratory markers most recent values

1. Perfusion/Oxygenation markers

- a. Lactate >2.5 mmol/L
- b. pH <7.35 or > 7.45
- c. SpO2 consistently <92% or Sao2 <90% on supplemental O2, or CVO2 <65

2. Coagulation Markers

- a. Elevated D-Dimer >500
- b. INR >1.5 or PTT elevated (not coagulant related)

3. Tissue Viability /Nutrition Markers

- a. Albumin <3.0
- b. Hemoglobin <8.0
- c. Hcrit <24%

Sub-Domain Scoring	Attributed Point Range
> or = to 5 abnormal markers across > or = to 2 categories:	3
3-4 abnormal markers	2
1-2 abnormal markers	1
0 abnormal markers	0
<i>**cluster bonus: if abnormal marks are present in ALL categories +1 for severity and significant supporting evidence</i>	
Total Tally (D2a)___/4	

- ii. Sub-Domain: Hemodynamic instability (current or recent) **0-4**
 - 1. Pressor Use (select one)
 - a. Currently on vasopressors
 - b. Recently discontinued vasopressors
 - c. History of vasopressor use during this admission but >72 hours ago
 - 2. Number of pressor agents (select one)
 - a. Single vasopressor
 - b. > or = to 2 vasopressors
 - 3. Vasopressor Dose Intensity
 - a. Low moderate dose (Ex: NorEPI < or = to 0.1 mcg/kg/min)
 - b. High dose or max dosing of any agent (Ex: NorEPI >0.1 mcg/kg/min)
 - 4. Vasopressor Duration (select one)
 - a. <24 hours
 - b. 24-72 hours
 - c. >72 hours
 - 5. Mean Arterial Pressor/MAP (select one)
 - a. MAP <60-sustained or recurrent, documented > or = 2 episodes
 - b. MAP 60-65 requiring active intervention to maintain result

Sub-Domain Scoring	Attributed Point Range
> or = to 2 pressors running concurrently and/or high dose and/or duration >72 hours AND MAP<60	4
Single pressor running at moderate dose >24 hours OR >= 2 running at ANY dose and duration.	3
Single vasopressor <24 hours OR discontinued within 72 hours OR MAP <60 without active pressors.	2
Hx of vasopressor use this admission <72 hours ago, OR borderline MAP (60-65) requiring intervention to maintain	1
No vasopressor use, and stable MAP >65.	0
Total sub-domain score (D2b)= __/4	
Total Domain Score (D2a+D2b)= __/8	

c. Domain 3: Wound Characteristics

i.Sub-Domain: Wound Location

1. Over bony prominence (typical pressure point)
2. Not over bony prominence (atypical pressure point)
3. Combination over and not over bony prominence (+1)
4. Anatomically Unilateral or bilateral (+1)
5. Over area of proposed angiosomes (+1)

ii.Sub-Domain: Wound Appearance:

1. Appears full thickness or partial thickness

2. Regular or irregular pattern with mottling radiating outward and poorly defined edges
3. Purple black maroon discoloration that is non blanching
4. Butterfly, pear shape, or horseshoe shape
5. Bruised appearance with linear striations
6. Regular oval round shape matching underlying bony prominence

iii.Sub-Domain: Wound Timeline (onset)

1. First documented >72 hours from point of admission? (may change with eCQM)
2. Developed in <24 hours from last intact skin assessment (extremely rapid)
3. Developed 24-72 hours (rapid)
4. Developed >72 hours (gradual)'

iv.Sub-Domain: Wound Timeline (progression)

1. Deteriorating within or after 2 weeks despite active medical treatment
2. stalling/staying the same, no change in 2 weeks despite active medical treatment
3. Improving within or after 2 weeks with active medical treatment (greater than 20% reduction in wound volume) (-1)
4. Recurrent necrosis following surgical debridement
5. Extending to new areas despite active medical treatment

d. Domain 4: Palliative Status / EOL Indicators

i.Hospice appropriate (<6-8 months life expectancy)

ii.DNR/DNI

iii.Palliative care team consulted

iv.Declining despite maximal medical therapy

v. Family/patient transitioning to comfort focused goals

Domain (1-4) Scoring

Domain	Max Attributive Points	% Of Determination
STOP-GAP	3	Modifier which contextualizes remainder of scoring
Domain 1: Clinical Context	7	28% weight
Domain 2: Hypoperfusion Markers	8	32% weight
Domain 3: Wound Characteristics	7	28% weight
Domain 4: EOL Markers	3	12% weight
Domain Tally:	25	100/100
Added STOP-GAP Tally	28	

****Finer points:**

** Domain 1+2= 60% of scoring, driving the determination of skin failure as a likely etiology.*

** Domain 3 accounts for ~ 30% of scoring, with expanded wound identification*

** Domain 4 accounts for ~ 12%, final determinant of skin failure classification (proceeding phases)*